

HACHINOHE, Japan

WMO: 475810

Lat: 40.5275N

Lon: 141.5219E

Elev: 29

StdP: 100.98

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-5.9	-12.0	1.3	-4.2	-10.9	1.5	-3.5	13.8	0.6	12.6	0.1	5.0	240	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	31.0	23.9	29.1	23.1	27.4	22.4	24.6	29.2	24.0	27.9	23.3	26.6	6.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.1	26.7	22.7	17.5	26.3	22.1	16.8	25.8	75.0	29.2	72.3	28.1	69.6	26.7	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	11.4	9.9	DB	-8.9	34.1	1.2	1.5	-9.7	35.2	-10.4	36.0	-11.1	36.8	-11.9	37.9	(4)
(5)			WB	-9.5	25.8	1.1	0.7	-10.3	26.3	-11.0	26.7	-11.6	27.2	-12.5	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-0.8	-0.2	3.1	8.7	13.8	16.9	21.2	23.1	19.7	13.6	7.3	1.8
	DBStd	8.80	2.51	2.89	3.29	3.72	3.65	3.17	3.56	3.14	3.02	3.00	3.58	2.93
	HDD10.0	1264	335	286	214	68	8	0	0	0	0	4	95	254
	HDD18.3	3103	593	519	471	289	146	63	12	2	18	149	330	512
	CDD10.0	1544	0	0	2	29	127	208	348	406	292	117	14	1
	CDD18.3	340	0	0	0	1	6	21	102	149	59	2	0	0
	CDH23.3	1957	0	0	0	6	46	94	601	947	260	4	0	0
	CDH26.7	520	0	0	0	1	7	16	169	274	53	0	0	0
Wind	WSAvg	4.5	5.4	5.3	5.5	5.2	4.3	3.4	3.2	3.4	3.7	4.3	4.9	5.3
Precipitation	PrecAvg	1104	57	47	61	64	88	104	132	133	166	106	74	60
	PrecMax	1444	137	170	120	163	160	235	277	322	322	223	162	164
	PrecMin	814	19	15	15	12	27	28	12	19	42	21	15	21
	PrecStd	158	30	31	28	36	38	55	67	83	71	62	41	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	11.5	17.2	23.8	27.2	28.9	32.8	33.6	30.9	23.7	19.4	13.0
		MCWB	5.0	6.7	10.3	13.6	16.1	19.7	24.4	24.8	23.9	17.8	13.9	9.6
	2%	DB	5.9	8.1	13.5	20.0	24.3	25.8	30.6	31.5	28.0	21.6	16.8	10.4
		MCWB	3.1	4.5	7.9	11.8	15.5	18.8	23.6	24.5	22.6	16.6	12.2	7.4
	5%	DB	4.4	6.2	10.9	17.2	22.1	23.9	28.5	29.7	26.1	20.2	15.0	8.5
		MCWB	1.9	2.9	6.5	10.3	14.7	18.2	22.9	23.8	21.5	15.8	11.2	5.8
	10%	DB	3.2	4.6	8.8	14.9	19.9	22.0	26.7	27.9	24.5	18.9	13.4	6.6
		MCWB	0.9	1.7	4.9	9.3	13.9	17.5	22.2	23.1	20.5	15.1	10.1	4.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	7.6	10.7	14.7	18.4	21.9	25.1	25.7	24.7	19.8	15.1	10.3
		MCDB	7.4	10.7	15.7	20.5	24.0	25.7	30.8	31.0	29.0	21.6	17.5	12.5
	2%	WB	3.5	5.0	8.6	12.8	17.0	20.3	24.2	25.0	23.6	17.9	13.0	7.9
		MCDB	5.5	7.8	12.7	18.2	21.6	24.1	29.0	29.7	26.9	20.1	15.9	10.1
	5%	WB	2.2	3.3	7.0	11.4	15.7	19.1	23.4	24.4	22.5	16.7	11.7	6.0
		MCDB	4.0	5.8	10.4	15.7	20.5	22.5	27.6	28.5	25.4	19.2	14.3	8.2
	10%	WB	1.2	1.9	5.3	10.0	14.5	18.0	22.7	23.7	21.2	15.6	10.5	4.3
		MCDB	2.8	4.2	8.4	14.3	18.8	21.4	26.2	27.3	23.6	18.2	13.0	6.4
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.7	7.7	8.9	8.5	6.9	6.4	6.2	7.2	8.5	8.0	6.4
		MCDBR	7.6	9.4	11.4	12.6	12.6	10.3	9.5	9.0	8.8	9.5	10.6	9.1
		MCWBR	5.8	6.8	7.5	6.9	6.0	4.8	4.1	3.6	4.5	5.7	7.2	7.0
	5% WB	MCDBR	7.5	9.2	10.9	11.1	10.9	9.0	8.8	8.0	7.9	7.7	9.6	9.0
MCWBR		6.1	7.1	7.9	7.0	6.1	4.8	4.1	3.5	4.3	5.4	7.3	7.4	
Clear-Sky Solar Irradiance	taub	0.366	0.410	0.481	0.530	0.545	0.526	0.528	0.500	0.444	0.416	0.397	0.376	
	taud	2.066	1.944	1.855	1.825	1.865	1.994	2.062	2.149	2.261	2.211	2.160	2.107	
	Ebn at Noon	750	770	757	753	754	768	763	772	793	765	715	703	
	Edh at Noon	115	151	185	203	200	176	163	146	122	115	105	103	
All-Sky Solar Radiation	RadAvg	1.61	2.32	3.40	4.45	4.92	4.82	4.30	4.12	3.57	2.80	1.86	1.37	
	RadStd	0.07	0.19	0.24	0.40	0.51	0.36	0.48	0.51	0.29	0.25	0.14	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	+0.47	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page